

BTS(C) – IV– 09 –013-G

B. Tech Degree IV Semester Examination, April 2009

CE 402 A/B SURVEYING II
(2006 Scheme)

Time: 3 Hours

Maximum Marks: 100

PART A

(5 x 8 = 40)

- I.
- What are the elements of a simple curve?
 - What is a simple curve? Mention the linear methods of setting out of simple curves.
 - What is the necessity of having various co-ordinate systems in field astronomy? Which all are the co-ordinate systems to locate the position of a heavenly body.
 - What do you understand by 'Echosounding'? Mention its advantages over older methods.
 - Give the factors for selection of Triangulation Stations.
 - What do you understand by satellite station and Reduction to centre?
 - Briefly explain types of errors.
 - State and prove the principle of least squares.

PART B

(4 x 15 = 60)

- II.
- Calculate the offsets at 10 m intervals for 5 points along the tangent to locate a curve having a radius of 200m. Adopt method of radial offsets. (5)
 - Two tangents intersect at chainage 59 + 60, the deflection angle being $50^{\circ}30'$. Calculate the necessary data for setting out the curve, if it is intended to set out the curve by Rankine's method of tangential angles. Take peg interval equal to 100 links, length of the chain being equal to 20 m. (10)
- OR**
- III.
- What are the elements of a compound curve? How will you set out it? Explain the procedure. (5)
 - CD and EF are two stations such that C & F are on opposite sides of a common tangent DE. It is required to connect CD & EF with a reverse curve, given that angles CDE and DEF are $151^{\circ}40'$ and $142^{\circ}20'$ respectively and that DE is 372.9m chainage of D is 2534.4m. Calculate the common radius and the chainage of the points of tangency and the point of reverse curvature. (10)
- IV.
- Observations were made from instrument station A to the signal at B. The sun makes an angle of 60° with the line BA. Calculate the phase correction of (i) the observation was made on the bright portion and (ii) the observation was made on the bright line. The distance AB is 9460 metres. The diameter of the signal is 12cm. (5)
 - The altitudes of two proposed stations A and B 100 km apart, are respectively 420m and 700 m. The intervening obstruction situated at C, 70 km from A has an elevation of 478m. Ascertain if A and B are intervisible and if necessary, find by how much B should be raised so that the line of sight must nowhere be less than 3m above the surface of the ground. (10)

OR

(Turn over)

- V. a. Explain 'laws of weights'. (5)
 b. The following are the three angles α, β and γ observed at a station P closing the horizon, along with their probable errors of measurement. Determine their corrected values.

$$\alpha = 78^\circ 12' 2'' \pm 2''$$

$$\beta = 136^\circ 48' 30'' \pm 4''$$

$$\gamma = 144^\circ 59' 08'' \pm 5'' \quad (10)$$

- VI. a. A star has a declination of $50^\circ 15' N$. Its upper (culmination) transit is in the Zenith of the place. Find the altitude of the star at the lower transit. (5)
 b. Find the azimuth and altitude of a star from the following data
 Latitude of the observer's place = $46^\circ N$
 Hour Angle of the star = $20^h 40^m$
 Declination = $18^\circ 38' S$ (10)

OR

- VII. The following observations were made on three stations A, B & C from a boat at O with the help of a sextant. Station B & O being on the same side of AC.

$$\angle AOB = 30^\circ 25' \quad \angle BOC = 45^\circ 25' \quad \angle ABC = 130^\circ 10'$$

$$AB = 4000 \text{ m} \quad BC = 4995 \text{ m}$$

Calculate the distance of the boat from the three stations. (15)

- VIII. a. Derive an expression for scale of vertical photograph. Explain how the ground Co-ordinates and the distance can be obtained from a vertical photograph. (10)
 b. Two consecutive photographs were taken with a camera of focal length 200 mm mounted on an airplane flying at a height of 1500m. The overlap was exactly 60% and the size of prints was 250 mm x 250mm. The flying height was same in the case of both the exposures and the flight was balanced so that there was no drift. The ground was flat and was 250m above the mean sea level. Determine the scale of the photograph and length of the airbase. (5)

OR

- IX. a. Explain:-
 (i) Equation of Time
 (ii) Mean Solar Time
 (iii) Sidereal Time (5)
 b. Find the L.S.T. corresponding to 5.05 a.m. on May 20, 1990 at a place on longitude $54^\circ 12' W$. The G.S.T. of G.M.M. being $7^h 12^m 54^s$. (10)

